Rigid-body motion notes

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1 Setting

- Center of mass:
$$\mathbf{x}_{\text{com}} = \frac{\int_{\Omega^t} \mathbf{x}\rho(x,t)\,d\mathbf{x}}{\text{mass}[\Omega^t]}.$$
where
$$\text{mass}[\Omega^t] = \int_{\Omega^t} \rho(x,t)\,d\mathbf{x}$$

- Rigid-body motion:
$$\mathbf{x} - \mathbf{x}_{\text{com}}(t) = \mathbf{Q}(t)(\mathbf{X} - \mathbf{X}_{\text{com}}) \Longleftrightarrow \mathbf{x} = \mathbf{x}_{\text{com}}(t) + \mathbf{Q}(t)(\mathbf{X} - \mathbf{X}_{\text{com}}),$$
where $\mathbf{Q}(t)$ is the rotation matrix, and $\mathbf{X}_{\text{com}}$ is the center of mass in the reference configuration. Correspondingly,
$$\mathbf{X} - \mathbf{X}_{\text{com}} = \mathbf{Q}^T(t)(\mathbf{x} - \mathbf{x}_{\text{com}}(t)) \Longleftrightarrow \mathbf{X} = \mathbf{X}_{\text{com}} + \mathbf{Q}^T(t)(\mathbf{x} - \mathbf{x}_{\text{com}}(t)).$$

- Linear momentum:
$$\mathbf{p}[\Omega^t] = \int_{\Omega^t} \rho(\mathbf{x},t)\mathbf{v}(\mathbf{x},t)\,d\mathbf{x}.$$

- Angular momentum:
$$\mathbf{L}[\Omega^t] = \int_{\Omega^t} \rho(\mathbf{x},t)(\mathbf{x} - \mathbf{x}_{\text{com}}) \times \mathbf{v}(\mathbf{x},t)\,d\mathbf{x}.$$

2 Rigid-body Motions

We want to find the angular velocity.

$$\begin{aligned}
\dot{\mathbf{x}} &= \mathbf{v}(\mathbf{x},t) \\
&= \dot{\mathbf{Q}}(t)\mathbf{X} + \dot{\mathbf{c}}(t) \\
&= \underbrace{\dot{\mathbf{Q}}(t)(\mathbf{Q}^T(t)}_{\boldsymbol{\omega}^\times}(\mathbf{x} - \mathbf{c}(t))) + \dot{\mathbf{c}}(t)
\end{aligned}$$

where $\boldsymbol{\omega}\times \equiv \dot{\mathbf{Q}}(t)\mathbf{Q}^T(t)$.

Remark 2.1. *Recall* $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$. *Then*
$$\dot{\mathbf{Q}}\mathbf{Q}^T + \mathbf{Q}\dot{\mathbf{Q}}^T = \underbrace{\mathbf{O}}_{\textit{zero tensor}} \implies \boldsymbol{\omega}\times + (\boldsymbol{\omega}\times)^T = \mathbf{O}.$$

Thus, $\boldsymbol{\omega}\times$ *is skew-symmetric.*

3 Center of Mass

We show that $\mathbf{c}(t) = \mathbf{x}_{\text{com}}$.

$$
\begin{aligned}
\underbrace{\int_{\Omega^t} \mathbf{x}\rho(\mathbf{x},t)\,d\mathbf{x}}_{\text{LHS: Expressed in reference configuration}} &= \mathbf{x}_{\text{com}} \cdot \text{mass}[\Omega^t], \\
\underbrace{\int_{\Omega^t} \mathbf{x}\rho(\mathbf{x},t)\,d\mathbf{x}}_{\int_{\Omega^0} \phi(\mathbf{X},t)\rho(\phi(\mathbf{X},t),t)J(\mathbf{X},t)\,d\mathbf{X}} &= \mathbf{x}_{\text{com}} \cdot \text{mass}[\Omega^t], \\
\implies \quad \int_{\Omega^0} \big(\mathbf{Q}(t)\mathbf{X} + \mathbf{c}(t)\big)\rho^0(\mathbf{X})\,d\mathbf{X} &= \mathbf{x}_{\text{com}} \cdot \text{mass}[\Omega^t], \\
\implies \quad \underbrace{\int_{\Omega^0} \mathbf{Q}(t)\mathbf{X}\rho^0(\mathbf{X})\,d\mathbf{X}}_{0} + \underbrace{\int_{\Omega^0} \mathbf{c}(t)\rho^0(\mathbf{X})\,d\mathbf{X}}_{\mathbf{c}(t)\int_{\Omega^0}\rho^0(\mathbf{X})\,d\mathbf{X}} &= \mathbf{x}_{\text{com}} \cdot \text{mass}[\Omega^t], \\
\implies \quad \mathbf{c}(t)\,\text{mass}[\Omega^0] &= \mathbf{x}_{\text{com}} \cdot \text{mass}[\Omega^t], \\
\implies \quad \mathbf{c}(t) &= \mathbf{x}_{\text{com}}.
\end{aligned}
$$

using the conservation of mass.

4 Linear Momentum

Recall that the velocity field of a rigid body is given by:

$$\mathbf{v}(\mathbf{x},t) = \boldsymbol{\omega}(t) \times (\mathbf{x} - \mathbf{c}(t)) + \dot{\mathbf{c}}(t),$$

and that the center of mass is located at $\mathbf{x}_{\text{com}} = \mathbf{c}(t)$.

The linear momentum of the rigid body is:

$$\mathbf{p}[\Omega^t] = \int_{\Omega^t} \rho(\mathbf{x},t)\,\mathbf{v}(\mathbf{x},t)\,d\mathbf{x}.$$

Substituting the velocity expression:

$$
\begin{aligned}
\mathbf{p}[\Omega^t] &= \int_{\Omega^t} \rho(\mathbf{x},t)\left[\boldsymbol{\omega}(t) \times (\mathbf{x} - \mathbf{c}(t)) + \dot{\mathbf{c}}(t)\right]\,d\mathbf{x} \\
&= \underbrace{\int_{\Omega^t} \rho(\mathbf{x},t)\left[\boldsymbol{\omega}(t) \times (\mathbf{x} - \mathbf{c}(t))\right]\,d\mathbf{x}}_{=0 \text{ (symmetry around the center of mass)}} + \dot{\mathbf{c}}(t)\int_{\Omega^t} \rho(\mathbf{x},t)\,d\mathbf{x}.
\end{aligned}
$$

The first term vanishes because $(\mathbf{x} - \mathbf{c}(t))$ is measured relative to the center of mass, and the integral is symmetric. The second term is simply the total mass:

$$\int_{\Omega^t} \rho(\mathbf{x},t)\,d\mathbf{x} = \text{mass}[\Omega^t].$$

Therefore, the linear momentum simplifies to:

$$\boxed{\mathbf{p}[\Omega^t] = \text{mass}[\Omega^t] \cdot \dot{\mathbf{c}}(t).}$$

This result shows that the linear momentum depends solely on the velocity of the center of mass $\dot{\mathbf{c}}(t)$ and the total mass $\text{mass}[\Omega^t]$.

5 Angular Momentum

Recall that the velocity field of a rigid body is:

$$\mathbf{v}(\mathbf{x},t) = \boldsymbol{\omega}(t) \times (\mathbf{x} - \mathbf{c}(t)) + \dot{\mathbf{c}}(t).$$

The angular momentum about the center of mass is:

$$\begin{aligned}\mathbf{L}[\Omega^t] &= \int_{\Omega^t} \rho(\mathbf{x},t)(\mathbf{x} - \mathbf{c}(t)) \times \mathbf{v}(\mathbf{x},t)\, d\mathbf{x} \\ &= \int_{\Omega^t} \rho(\mathbf{x},t)(\mathbf{x} - \mathbf{c}(t)) \times (\boldsymbol{\omega}(t) \times (\mathbf{x} - \mathbf{c}(t)))\ d\mathbf{x} \quad + \underbrace{\int_{\Omega^t} \rho(\mathbf{x},t)(\mathbf{x} - \mathbf{c}(t))\, d\mathbf{x}}_{=0} \times \dot{\mathbf{c}}(t).\end{aligned}$$

The second term vanishes due to symmetry around the center of mass. For the first term, define the cross-product operator:

$$\mathbf{R}(\mathbf{x},t) \equiv (\mathbf{x} - \mathbf{c}(t))\times,$$

so we rewrite:

$$\mathbf{L}[\Omega^t] = \int_{\Omega^t} \rho(\mathbf{x},t)\, \mathbf{R}(\mathbf{x},t)\, \mathbf{R}^\top(\mathbf{x},t)\, d\mathbf{x}\, \boldsymbol{\omega}(t).$$

Define the spatial inertia tensor:

$$\mathbf{I}[\Omega^t] \equiv \int_{\Omega^t} \rho(\mathbf{x},t)\, \mathbf{R}(\mathbf{x},t)\, \mathbf{R}^\top(\mathbf{x},t)\, d\mathbf{x}.$$

Therefore, the angular momentum is:

$$\boxed{\mathbf{L}[\Omega^t] = \mathbf{I}[\Omega^t]\boldsymbol{\omega}(t).}$$

This expression shows that angular momentum is the product of the inertia tensor (with respect to the center of mass) and the angular velocity vector.

6 Torque

6.1 Inertia Tensor Transformation

We first show that $\mathbf{I}[\Omega^t] = \mathbf{Q}(t)\mathbf{I}[\Omega^0]\mathbf{Q}^T(t)$.

Recall that $\mathbf{I}[\Omega^t] = \int_{\Omega^t} \rho(\mathbf{x},t)\mathbf{R}(\mathbf{x},t)\mathbf{R}^T(\mathbf{x},t)\, d\mathbf{x}$ is the inertia tensor. With $\mathbf{x} = \phi(\mathbf{X},t) = \mathbf{Q}(t)(\mathbf{X} - \mathbf{X}_{\text{com}}) + \mathbf{c}(t)$, it follows that $(\mathbf{x} - \mathbf{c}(t))\times = \mathbf{Q}(t)(\mathbf{X} - \mathbf{X}_{\text{com}})\times$.

The skew-symmetric matrix $\mathbf{R}$ is given by:

$$\mathbf{R} = \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix}, \quad \text{where } r_i = x_i - c_i \text{ is the } i\text{-th entry of } \mathbf{r}.$$

Then:

$$\mathbf{R}\mathbf{R}^T = \begin{bmatrix} r_2^2 + r_3^2 & -r_1 r_2 & -r_1 r_3 \\ -r_1 r_2 & r_1^2 + r_3^2 & -r_2 r_3 \\ -r_1 r_3 & -r_2 r_3 & r_1^2 + r_2^2 \end{bmatrix} = \|\mathbf{r}\|^2 \mathbf{I} - \mathbf{r}\mathbf{r}^T.$$

Substituting $\mathbf{r} = \mathbf{x} - \mathbf{c}(t)$:

$$\mathbf{R}\mathbf{R}^T = \|\mathbf{x} - \mathbf{c}(t)\|^2 \mathbf{I} - (\mathbf{x} - \mathbf{c}(t))(\mathbf{x} - \mathbf{c}(t))^T.$$

Using $\mathbf{x} - \mathbf{c}(t) = \mathbf{Q}(t)(\mathbf{X} - \mathbf{X}_{\text{com}})$:

$$\mathbf{R}\mathbf{R}^T = \|\mathbf{Q}(t)(\mathbf{X} - \mathbf{X}_{\text{com}})\|^2 \mathbf{Q}\mathbf{Q}^T - \mathbf{Q}(\mathbf{X} - \mathbf{X}_{\text{com}})(\mathbf{X} - \mathbf{X}_{\text{com}})^T \mathbf{Q}^T.$$

Simplify:

$$\mathbf{R}\mathbf{R}^T = \mathbf{Q}(t) \left[\|\mathbf{X} - \mathbf{X}_{\text{com}}\|^2 \mathbf{I} - (\mathbf{X} - \mathbf{X}_{\text{com}})(\mathbf{X} - \mathbf{X}_{\text{com}})^T \right] \mathbf{Q}^T.$$

Thus:

$$\mathbf{R}\mathbf{R}^T = \mathbf{Q}(t)\mathbf{R}_0\mathbf{Q}^T(t), \quad \text{where } \mathbf{R}_0 = (\mathbf{X} - \mathbf{X}_{\text{com}}) \times (\mathbf{X} - \mathbf{X}_{\text{com}}).$$

Finally, compute $\mathbf{I}[\Omega^t]$:

$$\mathbf{I}[\Omega^t] = \int_{\Omega^t} \rho(\mathbf{x}, t)\mathbf{R}(\mathbf{x}, t)\mathbf{R}^T(\mathbf{x}, t)\, d\mathbf{x}.$$

Substituting $\mathbf{x} = \phi(\mathbf{X}, t)$:

$$\mathbf{I}[\Omega^t] = \int_{\Omega^0} \rho^0(\mathbf{X}, t)\mathbf{Q}(t)\mathbf{R}_0\mathbf{Q}^T(t)\, d\mathbf{X}.$$

Factor out $\mathbf{Q}(t)$:

$$\mathbf{I}[\Omega^t] = \mathbf{Q}(t) \underbrace{\int_{\Omega^0} \rho^0(\mathbf{X}, t)\mathbf{R}_0\, d\mathbf{X}}_{\mathbf{I}[\Omega^0]} \mathbf{Q}^T(t).$$

Recognizing $\mathbf{I}[\Omega^0]$ as the reference inertia tensor:

$$\boxed{\mathbf{I}[\Omega^t] = \mathbf{Q}(t)\mathbf{I}[\Omega^0]\mathbf{Q}^T(t).}$$

Remark 6.1. *In practical implementations, it is often convenient to work with a discrete approximation of the inertia tensor. Recall that in the continuous setting, the inertia tensor in the reference configuration Ω_0 is defined as:*

$$\mathbf{I}[\Omega_0] = \int_{\Omega_0} \rho^0(\mathbf{X}) \left(\|\mathbf{X} - \mathbf{X}_{com}\|^2 \mathbf{I}_d - (\mathbf{X} - \mathbf{X}_{com})(\mathbf{X} - \mathbf{X}_{com})^T \right) d\mathbf{X}.$$

To discretize this expression, we replace the integral with a finite sum over point masses m_i located at positions $\mathbf{X}_i$, and define $\hat{\mathbf{r}}_i := \mathbf{X}_i - \mathbf{X}_{com}$. Then:

$$\mathbf{I}[\Omega_0] \approx \sum_i m_i \left(\|\hat{\mathbf{r}}_i\|^2 \mathbf{I}_d - \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^T \right).$$

Now define the matrix:

$$\hat{\mathbf{I}} := \sum_i m_i\, \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^T,$$

which is the second-moment matrix of the point mass positions relative to the center of mass. Then we note that:

$$\sum_i m_i \|\hat{\mathbf{r}}_i\|^2 = \sum_i m_i \operatorname{Tr}(\hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^T) = \operatorname{Tr}\left(\sum_i m_i \hat{\mathbf{r}}_i \hat{\mathbf{r}}_i^T \right) = \operatorname{Tr}(\hat{\mathbf{I}}).$$

Substituting this into the discrete inertia formula gives:

$$\mathbf{I}[\Omega_0] = \operatorname{Tr}(\hat{\mathbf{I}})\, \mathbf{I}_d - \hat{\mathbf{I}}.$$

This expression avoids explicit cross products and is particularly efficient when the body is represented by discrete mass samples. It also makes numerical implementation more convenient in practice.

6.2 Torque

Recall that

$$\mathbf{I}[\Omega^t] = \int_{\Omega^t} \rho(\mathbf{x},t)\mathbf{R}\mathbf{R}^T \, d\mathbf{x} = \int_{\Omega^0} \rho^0(\mathbf{X},t)\mathbf{R}(\phi(\mathbf{X},t),t)\mathbf{R}^T(\phi(\mathbf{X},t),t) \, d\mathbf{X}.$$

Using the relationship $(\mathbf{x}-\mathbf{c}(t))\times = \mathbf{Q}(t)(\mathbf{X}-\mathbf{X}_{\text{com}})\times$, we find:

$$(\mathbf{X}-\mathbf{X}_{\text{com}})\times = \mathbf{Q}^T(t)(\mathbf{x}-\mathbf{c}(t)) \times .$$

The time derivative of $\mathbf{R}$ is:

$$\frac{d\mathbf{R}}{dt} = (\mathbf{x}-\mathbf{c}(t))\times = \dot{\mathbf{Q}}(t)[\mathbf{X}-\mathbf{X}_{\text{com}}]\times = \dot{\mathbf{Q}}\mathbf{Q}^T\mathbf{R} = \boldsymbol{\omega}^\times\mathbf{R}.$$

The derivative of $\mathbf{R}\mathbf{R}^T$ becomes:

$$\frac{d}{dt}(\mathbf{R}\mathbf{R}^T) = \dot{\mathbf{R}}\mathbf{R}^T + \mathbf{R}\dot{\mathbf{R}}^T = (\boldsymbol{\omega}^\times\mathbf{R})\mathbf{R}^T + \mathbf{R}(\boldsymbol{\omega}^\times\mathbf{R})^T.$$

Simplify the transpose term:

$$(\boldsymbol{\omega}^\times\mathbf{R})^T = \mathbf{R}^T(\boldsymbol{\omega}^\times)^T.$$

Thus:

$$\frac{d}{dt}(\mathbf{R}\mathbf{R}^T) = (\boldsymbol{\omega}^\times\mathbf{R})\mathbf{R}^T - \mathbf{R}(\mathbf{R}^T\boldsymbol{\omega}^\times).$$

The torque, which is the derivative of angular momentum, is given by:

$$\tau[\Omega^t] = \frac{d}{dt}(\mathbf{I}[\Omega^t]\boldsymbol{\omega}).$$

Expanding this:

$$\tau[\Omega^t] = \int_{\Omega^0} \rho^0(\mathbf{X})\frac{d}{dt}(\mathbf{R}\mathbf{R}^T) \, d\mathbf{X}\, \boldsymbol{\omega} + \mathbf{I}[\Omega^t]\dot{\boldsymbol{\omega}}.$$

Substitute $\frac{d}{dt}(\mathbf{R}\mathbf{R}^T)$:

$$\tau[\Omega^t] = \int_{\Omega^0} \rho^0(\mathbf{X})[(\boldsymbol{\omega}^\times\mathbf{R})\mathbf{R}^T - \mathbf{R}(\mathbf{R}^T\boldsymbol{\omega}^\times)] \, d\mathbf{X}\, \boldsymbol{\omega} + \mathbf{I}[\Omega^t]\dot{\boldsymbol{\omega}}.$$

Distribute the integral:

$$\tau[\Omega^t] = \boldsymbol{\omega} \times \mathbf{I}[\Omega^t]\boldsymbol{\omega} - \mathbf{I}[\Omega^t](\boldsymbol{\omega} \times \boldsymbol{\omega}) + \mathbf{I}[\Omega^t]\dot{\boldsymbol{\omega}}.$$

The cross-product of a vector with itself is zero:

$$\boldsymbol{\omega} \times \boldsymbol{\omega} = 0.$$

After simplification:

$$\boxed{\tau[\Omega^t] = \boldsymbol{\omega} \times \mathbf{I}[\Omega^t]\boldsymbol{\omega} + \mathbf{I}[\Omega^t]\dot{\boldsymbol{\omega}}.}$$

Here, the cross-product term $\boldsymbol{\omega} \times \mathbf{I}[\Omega^t]\boldsymbol{\omega}$ represents the torque due to angular momentum coupling, while $\mathbf{I}[\Omega^t]\dot{\boldsymbol{\omega}}$ accounts for the angular acceleration contribution.

6.3 Constant Angular Velocity for a Free Rigid Body

Assume that for a free rigid body,

$$\boldsymbol{\omega} = \|\boldsymbol{\omega}\| \, \boldsymbol{q}_k$$

where $\boldsymbol{q}_k$ is an eigenvector of the inertia tensor $\mathbf{I}[\Omega^t]$ such that

$$\mathbf{I}[\Omega^t]\boldsymbol{q}_k = \lambda_k \boldsymbol{q}_k.$$

The equation of motion for angular momentum states:

$$\boldsymbol{\tau} = \boldsymbol{\omega} \times \mathbf{I}[\Omega^t]\boldsymbol{\omega} + \mathbf{I}\dot{\boldsymbol{\omega}} = 0.$$

Since $\boldsymbol{\omega} = \|\boldsymbol{\omega}\| \, \boldsymbol{q}_k$, we substitute this into the cross-product term:

$$\boldsymbol{\omega} \times \mathbf{I}[\Omega^t]\boldsymbol{\omega} = (\|\boldsymbol{\omega}\|^2 \, \lambda_k)(\boldsymbol{q}_k \times \boldsymbol{q}_k) = 0.$$

Thus,

$$\mathbf{I}\dot{\boldsymbol{\omega}} = 0 \implies \dot{\boldsymbol{\omega}} = 0.$$

Therefore, the angular velocity remains constant over time:

$$\boxed{\boldsymbol{\omega}(t) = \boldsymbol{\omega}(0), \quad \forall t.}$$

7 The Principle of Least Action in One Dimension

Before considering rigid-body motion, we illustrate the variational principle in a simple setting: a particle of mass m attached to a linear spring with stiffness k, moving along a line.

Action and Energy

The kinetic and potential energy are given by:

$$\mathrm{KE} = \frac{1}{2}m\dot{x}(t)^2, \qquad \mathrm{PE} = \frac{1}{2}kx(t)^2.$$

The action functional is:

$$S[x(t)] = \int_0^T (\mathrm{KE} - \mathrm{PE}) \, dt = \int_0^T \left(\frac{1}{2}m\dot{x}(t)^2 - \frac{1}{2}kx(t)^2 \right) dt.$$

Variation of the Action

Consider a variation of the trajectory:

$$x_\varepsilon(t) = x(t) + \varepsilon \, \delta x(t),$$

with $\delta x(0) = \delta x(T) = 0$.

The variation of the action is:

$$\delta S = \frac{d}{d\varepsilon} S[x_\varepsilon(t)]\bigg|_{\varepsilon=0} = \int_0^T \Big(m\dot{x}(t)\dot{\delta x}(t) - kx(t)\delta x(t) \Big) \, dt.$$

Apply integration by parts to the first term:

$$\int_0^T m\dot{x}(t)\dot{\delta x}(t)dt = -\int_0^T m\ddot{x}(t)\delta x(t)dt,$$

where boundary terms vanish due to $\delta x(0) = \delta x(T) = 0$.

Substituting back gives:

$$\delta S = \int_0^T \left(-m\ddot{x}(t) - kx(t)\right)\delta x(t)\, dt.$$

Euler–Lagrange Equation

Since $\delta x(t)$ is arbitrary, the integrand must vanish:

$$\boxed{m\ddot{x}(t) + kx(t) = 0.}$$

This is the classical equation of motion for a simple harmonic oscillator. The variational structure of this scalar system mirrors the full rigid-body case developed in later sections.

8 Kinetic Energy in Rigid-Body Motion

The kinetic energy of the rigid body is given by:

$$KE = \frac{1}{2}\int_{\Omega^0} \rho^0(\mathbf{X})\mathbf{v}^T(\boldsymbol{\phi}(\mathbf{X},t),t)\mathbf{v}(\boldsymbol{\phi}(\mathbf{X},t),t)\, d\mathbf{X},$$

where the velocity $\mathbf{v}(\boldsymbol{\phi}(\mathbf{X},t),t)$ is:

$$\mathbf{v}(\boldsymbol{\phi}(\mathbf{X},t),t) = \boldsymbol{\omega} \times \mathbf{Q}(t)(\mathbf{X} - \mathbf{X}_{\text{com}}) + \dot{\mathbf{c}}(t).$$

Let $\mathbf{r} = \mathbf{Q}(t)(\mathbf{X} - \mathbf{X}_{\text{com}})$. Then:

$$\mathbf{v}^T\mathbf{v} = (\boldsymbol{\omega} \times \mathbf{r} + \dot{\mathbf{c}}(t))^T(\boldsymbol{\omega} \times \mathbf{r} + \dot{\mathbf{c}}(t)).$$

Expanding:

$$\mathbf{v}^T\mathbf{v} = (\boldsymbol{\omega} \times \mathbf{r})^T(\boldsymbol{\omega} \times \mathbf{r}) + 2(\boldsymbol{\omega} \times \mathbf{r})^T\dot{\mathbf{c}}(t) + \|\dot{\mathbf{c}}(t)\|^2.$$

Contribution of $\|\dot{\mathbf{c}}(t)\|^2$:

$$\int_{\Omega^0} \rho^0(\mathbf{X})\|\dot{\mathbf{c}}(t)\|^2\, d\mathbf{X} = \|\dot{\mathbf{c}}(t)\|^2 \cdot \text{mass}[\Omega^0].$$

Contribution of $(\boldsymbol{\omega} \times \mathbf{r})^T\dot{\mathbf{c}}(t)$:

$$2\int_{\Omega^0} \rho^0(\mathbf{X})(\boldsymbol{\omega} \times \mathbf{r})^T\dot{\mathbf{c}}(t)\, d\mathbf{X} = 0,$$

because:

$$\int_{\Omega^0} \rho^0(\mathbf{X})(\mathbf{X} - \mathbf{X}_{\text{com}})\, d\mathbf{X} = \mathbf{0}.$$

Contribution of $(\boldsymbol{\omega} \times \mathbf{r})^T(\boldsymbol{\omega} \times \mathbf{r})$:

$$(\boldsymbol{\omega} \times \mathbf{r})^T(\boldsymbol{\omega} \times \mathbf{r}) = (\mathbf{r} \times \boldsymbol{\omega})^T(\mathbf{r} \times \boldsymbol{\omega}) = \boldsymbol{\omega}^T(\mathbf{r}\times)(\mathbf{r}\times)^T\boldsymbol{\omega}.$$

Using the property of $\mathbf{r}\times$ as a skew-symmetric matrix:

$$\int_{\Omega^0} \rho^0(\mathbf{X})(\boldsymbol{\omega} \times \mathbf{r})^T(\boldsymbol{\omega} \times \mathbf{r})\, d\mathbf{X} = \boldsymbol{\omega}^T \mathbf{I}[\Omega_t]\boldsymbol{\omega},$$

where $\mathbf{I}[\Omega_t] = \mathbf{Q}(t)\,\mathbf{I}[\Omega^0]\,\mathbf{Q}^T(t)$.

Combining the terms, the total kinetic energy is:

$$\boxed{KE = \underbrace{\frac{1}{2}\boldsymbol{\omega}^T \mathbf{I}[\Omega_t]\boldsymbol{\omega}}_{\text{rotational KE}} + \underbrace{\frac{1}{2}\|\dot{\mathbf{c}}(t)\|^2 \cdot \text{mass}[\Omega^0]}_{\text{translational KE}}.}$$

9 Variational Principle for Rigid-Body Motion

We derive the equations of motion for a rigid body using Hamilton's principle, which states that the physical trajectory extremizes the action functional. The total motion is decomposed into translation of the center of mass and rotation about the center of mass.

9.1 Lagrangian and Action Functional

The action is defined as:

$$S = \int_0^T L\, dt,$$

where the Lagrangian is the difference between kinetic and potential energy:

$$L = \text{KE} - \text{PE}.$$

For a freely moving rigid body with no external potential:

$$\text{PE} = 0.$$

The total kinetic energy consists of translational and rotational parts. It may be expressed in either the spatial or material frame.

Spatial Frame

$$\text{KE} = \frac{1}{2}\,\text{mass}[\Omega_0]\|\dot{\mathbf{c}}(t)\|^2 + \frac{1}{2}\boldsymbol{\omega}(t)^\top \mathbf{I}[\Omega_t]\boldsymbol{\omega}(t),$$

Material (Body) Frame

$$\text{KE} = \frac{1}{2}\,\text{mass}[\Omega_0]\|\dot{\mathbf{c}}(t)\|^2 + \frac{1}{2}\boldsymbol{\Omega}(t)^\top \mathbf{I}[\Omega_0]\boldsymbol{\Omega}(t),$$

with $\boldsymbol{\omega} = Q(t)\boldsymbol{\Omega}$, and $\mathbf{I}[\Omega_t] = Q(t)\mathbf{I}[\Omega_0]Q(t)^\top$.

9.2 Conservation of Linear Momentum

To derive the conservation of linear momentum, consider a variation of the center of mass trajectory:

$$\mathbf{c}_\varepsilon(t) = \mathbf{c}(t) + \varepsilon\, \delta\mathbf{c}(t),$$

where the variation $\delta\mathbf{c}(t)$ satisfies the boundary conditions

$$\delta\mathbf{c}(0) = \delta\mathbf{c}(T) = \mathbf{0}.$$

The time derivative of the perturbed trajectory is

$$\dot{\mathbf{c}}_\varepsilon(t) = \dot{\mathbf{c}}(t) + \varepsilon\, \dot{\delta\mathbf{c}}(t).$$

Then the action becomes

$$S[\mathbf{c}(t)] = \int_0^T \left(\frac{1}{2}\,\mathrm{mass}[\Omega_0]\, \|\dot{\mathbf{c}}(t)\|^2 + \frac{1}{2}\boldsymbol{\omega}^T \mathbf{I}[\Omega_t]\boldsymbol{\omega} \right) dt.$$

Under the variation $\mathbf{c}_\varepsilon(t) = \mathbf{c}(t) + \varepsilon\delta\mathbf{c}(t)$, the action becomes

$$S[\mathbf{c}_\varepsilon(t)] = \int_0^T \left(\frac{1}{2}\,\mathrm{mass}[\Omega_0]\, \|\dot{\mathbf{c}}_\varepsilon(t)\|^2 + \frac{1}{2}\boldsymbol{\omega}^T \mathbf{I}[\Omega_t]\boldsymbol{\omega} \right) dt.$$

Taking the derivative of the action with respect to ε:

$$\frac{d}{d\varepsilon} S[\mathbf{c}_\varepsilon(t)]\bigg|_{\varepsilon=0} = \mathrm{mass}[\Omega_0] \int_0^T \dot{\mathbf{c}}(t)^T \dot{\delta\mathbf{c}}(t)\, dt.$$

Now apply integration by parts:

$$\int_0^T \dot{\mathbf{c}}(t)^T \dot{\delta\mathbf{c}}(t)\, dt = \dot{\mathbf{c}}(t)^T \delta\mathbf{c}(t)\Big|_0^T - \int_0^T \ddot{\mathbf{c}}(t)^T \delta\mathbf{c}(t)\, dt.$$

Using the boundary conditions $\delta\mathbf{c}(0) = \delta\mathbf{c}(T) = \mathbf{0}$, the boundary term vanishes, and we are left with:

$$\frac{d}{d\varepsilon} S[\mathbf{c}_\varepsilon(t)]\bigg|_{\varepsilon=0} = -\,\mathrm{mass}[\Omega_0] \int_0^T \ddot{\mathbf{c}}(t)^\top \delta\mathbf{c}(t)\, dt.$$

Since $\delta\mathbf{c}(t)$ is arbitrary and vanishes at the endpoints, the integrand must vanish for all t, which yields the equation of motion:

$$\boxed{\frac{d}{dt}\left(\mathrm{mass}[\Omega_0]\, \dot{\mathbf{c}}(t)\right) = 0 \quad \Longleftrightarrow \quad \mathrm{mass}[\Omega_0]\, \ddot{\mathbf{c}}(t) = 0.}$$

9.3 Conservation of Angular Momentum

We now derive angular momentum conservation from a variational principle using quaternions and material-frame angular velocity $\boldsymbol{\Omega}(t) \in \mathbb{R}^3$.

Quaternion Variation

Let $q(t) \in \mathbb{H}$ be the unit quaternion representing the rigid-body orientation. We introduce a perturbation by defining a one-parameter family of quaternions:

$$q(t,\varepsilon) = q(t) \cdot \exp\left(\frac{\varepsilon}{2}(0, \delta\mathbf{q}(t))\right),$$

where $\delta\mathbf{q}(t) \in \mathbb{R}^3$ is a time-dependent perturbation vector and $(0, \delta\mathbf{q}) \in \mathbb{H}$ is a pure quaternion.

The quaternion kinematic equation under material angular velocity is:

$$\frac{dq}{dt}(t,\varepsilon) = q(t,\varepsilon) \cdot \frac{1}{2}(0, \mathbf{\Omega}(t,\varepsilon)).$$

Computing the Variation of Angular Velocity

Differentiating with respect to ε at $\varepsilon = 0$ gives:

$$\left.\frac{\partial q}{\partial \varepsilon}(t,\varepsilon)\right|_{\varepsilon=0} = q(t) \cdot \frac{1}{2}(0, \delta\mathbf{q}(t)).$$

Then compute the mixed derivatives:

$$\left.\frac{\partial^2 q}{\partial \varepsilon\, \partial t}\right|_{\varepsilon=0} = \dot{q}(t) \cdot \frac{1}{2}(0, \delta\mathbf{q}) + q(t) \cdot \frac{1}{2}(0, \dot{\delta\mathbf{q}}),$$

$$\left.\frac{\partial^2 q}{\partial t\, \partial \varepsilon}\right|_{\varepsilon=0} = \frac{d}{d\varepsilon}\left(q(t,\varepsilon) \cdot \frac{1}{2}(0, \mathbf{\Omega}(t,\varepsilon))\right)\Bigg|_{\varepsilon=0} = \frac{\partial q}{\partial \varepsilon} \cdot \frac{1}{2}(0, \mathbf{\Omega}) + q(t) \cdot \frac{1}{2}(0, \partial_\varepsilon \mathbf{\Omega}(t,0)).$$

Equating these and simplifying:

$$\dot{q} \cdot \frac{1}{2}(0, \delta\mathbf{q}) + q \cdot \frac{1}{2}(0, \dot{\delta\mathbf{q}}) = q \cdot \frac{1}{2}(0, \partial_\varepsilon \mathbf{\Omega}) + \left(q \cdot \frac{1}{2}(0, \delta\mathbf{q})\right) \cdot \frac{1}{2}(0, \mathbf{\Omega}).$$

Using the quaternion identity:

$$(0, \mathbf{a}) \cdot (0, \mathbf{b}) = (-\mathbf{a} \cdot \mathbf{b},\ \mathbf{a} \times \mathbf{b}),$$

we find:

$$(0, \mathbf{\Omega}) \cdot (0, \delta\mathbf{q}) - (0, \delta\mathbf{q}) \cdot (0, \mathbf{\Omega}) = 2(0, \mathbf{\Omega} \times \delta\mathbf{q}).$$

Thus:

$$(0, \partial_\varepsilon \mathbf{\Omega}) = \left(0, \dot{\delta\mathbf{q}} + \mathbf{\Omega} \times \delta\mathbf{q}\right), \quad \text{so} \quad \boxed{\delta\mathbf{\Omega} = \dot{\delta\mathbf{q}} + \mathbf{\Omega} \times \delta\mathbf{q}}.$$

Variation of the Action

Now vary the action:

$$\delta S = \int_0^T \mathbf{\Omega}^\top I[\Omega_0]\delta\mathbf{\Omega}\, dt = \int_0^T \mathbf{\Omega}^\top I[\Omega_0](\dot{\delta\mathbf{q}} + \mathbf{\Omega} \times \delta\mathbf{q})\, dt.$$

Integrate the first term by parts (boundary terms vanish):

$$\int_0^T \mathbf{\Omega}^\top I[\Omega_0]\dot{\delta\mathbf{q}}\, dt = -\int_0^T \dot{\mathbf{\Omega}}^\top I[\Omega_0]\delta\mathbf{q}\, dt.$$

The second term becomes:

$$\int_0^T (\boldsymbol{\Omega} \times I[\Omega_0]\boldsymbol{\Omega})^\top \, \delta\mathbf{q} \, dt.$$

Therefore:

$$\delta S = \int_0^T \left(-I[\Omega_0]\dot{\boldsymbol{\Omega}} + \boldsymbol{\Omega} \times I[\Omega_0]\boldsymbol{\Omega}\right)^\top \delta\mathbf{q} \, dt.$$

Since $\delta\mathbf{q}(t)$ is arbitrary:

$$\boxed{I[\Omega_0]\dot{\boldsymbol{\Omega}} + \boldsymbol{\Omega} \times I[\Omega_0]\boldsymbol{\Omega} = 0.}$$

Transformation to the Spatial Frame To express this in world coordinates, recall:

$$\boldsymbol{\omega} = Q\boldsymbol{\Omega}, \quad I[\Omega_t] = QI[\Omega_0]Q^\top.$$

Differentiate $\boldsymbol{\omega} = Q\boldsymbol{\Omega}$:

$$\dot{\boldsymbol{\omega}} = \dot{Q}\boldsymbol{\Omega} + Q\dot{\boldsymbol{\Omega}} = \boldsymbol{\omega}^\times\boldsymbol{\omega} + Q\dot{\boldsymbol{\Omega}}.$$

Multiply the body-frame equation by Q:

$$QI[\Omega_0]\dot{\boldsymbol{\Omega}} + Q(\boldsymbol{\Omega} \times I[\Omega_0]\boldsymbol{\Omega}) = 0.$$

Express in spatial quantities:

$$I[\Omega_t]Q\dot{\boldsymbol{\Omega}} + \boldsymbol{\omega} \times I[\Omega_t]\boldsymbol{\omega} = 0.$$

Substitute $Q\dot{\boldsymbol{\Omega}} = \dot{\boldsymbol{\omega}} - \boldsymbol{\omega}^\times\boldsymbol{\omega}$, and use $\boldsymbol{\omega} \times \boldsymbol{\omega} = 0$, to obtain:

$$\boxed{I[\Omega_t]\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times I[\Omega_t]\boldsymbol{\omega} = 0.}$$

This is the angular momentum conservation law in the spatial (world) frame.

10 Quaternions and Angular Velocity

Unit quaternions $q(t) \in \mathbb{H}$ are a compact and non-singular representation of 3D rotations. They are widely used in rigid-body dynamics due to their numerical stability and geometric clarity.

Let a quaternion be written as:

$$q = (q_0, \mathbf{q}) = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k},$$

where $q_0 \in \mathbb{R}$ is the scalar part and $\mathbf{q} \in \mathbb{R}^3$ is the vector part.

Quaternion Multiplication Given two quaternions $p = (p_0, \mathbf{p})$, $q = (q_0, \mathbf{q}) \in \mathbb{H}$, their product is:

$$p \cdot q = (p_0 q_0 - \mathbf{p} \cdot \mathbf{q},\ p_0\mathbf{q} + q_0\mathbf{p} + \mathbf{p} \times \mathbf{q}),$$

where $\mathbf{p}, \mathbf{q} \in \mathbb{R}^3$ and the result is another quaternion.

Equivalently, using component notation with the quaternion basis $\{e_0 = 1,\ e_1 = \mathbf{i},\ e_2 = \mathbf{j},\ e_3 = \mathbf{k}\}$, we define a rank-3 tensor $Q_{\alpha\beta\gamma} \in \mathbb{R}^{4\times4\times4}$ such that:

$$(p \cdot q)_\alpha = Q_{\alpha\beta\gamma}\, p_\beta\, q_\gamma.$$

The nonzero components of $Q_{\alpha\beta\gamma}$ are compactly encoded as a list of 4×4 matrices, one for each value of $\alpha \in \{0, 1, 2, 3\}$:

$$Q_{\alpha\beta\gamma} = \left[\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \right]$$

Each 4×4 matrix corresponds to the slice $Q_{\alpha\beta\gamma}$ for a fixed α. The first matrix gives $Q_{0\beta\gamma}$, the second gives $Q_{1\beta\gamma}$, and so on.

Note that quaternion multiplication is noncommutative:

$$p \cdot q \neq q \cdot p,$$

and quaternion multiplication is associative:

$$p \cdot (q \cdot r) = (p \cdot q) \cdot r.$$

Conjugate and Inverse

$$q^* = (q_0, -\mathbf{q}), \quad q^{-1} = \frac{q^*}{\|q\|^2}, \quad \text{and for unit quaternions: } q^{-1} = q^*.$$

Rotating a Vector To rotate a vector $\mathbf{v} \in \mathbb{R}^3$, write it as a pure quaternion $(0, \mathbf{v})$, and apply:

$$\mathbf{v}_{\text{rotated}} = \operatorname{Im}\left(q \cdot (0, \mathbf{v}) \cdot q^{-1}\right).$$

Angular Velocity from Quaternion Derivatives Let $q(t)$ be a unit quaternion representing the time-dependent rotation of a rigid body.

There are two angular velocity vectors associated with $q(t)$:

- **Material (Body) Frame Angular Velocity $\boldsymbol{\Omega}(t) \in \mathbb{R}^3$:** the angular velocity expressed in the body frame.

  In this case, the quaternion derivative satisfies:

  $$\dot{q}(t) = \frac{1}{2} q(t) \cdot (0, \boldsymbol{\Omega}(t)),$$

  and therefore,

  $$\boxed{\boldsymbol{\Omega}(t) = 2 \cdot \operatorname{Im}\left(q(t)^{-1} \dot{q}(t)\right)}.$$

- **Spatial (World) Frame Angular Velocity $\boldsymbol{\omega}(t) \in \mathbb{R}^3$:** the angular velocity expressed in the inertial frame.

  In this case:

  $$\dot{q}(t) = \frac{1}{2} (0, \boldsymbol{\omega}(t)) \cdot q(t),$$

  which leads to:

  $$\boxed{\boldsymbol{\omega}(t) = 2 \cdot \operatorname{Im}\left(\dot{q}(t) \cdot q(t)^{-1}\right)}.$$

Quaternion Time Integration and Frame Transformations

To numerically update the orientation of a rigid body, we represent rotation using unit quaternions $q(t) \in \mathbb{H}$, and angular velocity using a 3-vector $\boldsymbol{\omega}(t) \in \mathbb{R}^3$ in the spatial (world) frame.

Assuming the angular velocity is approximately constant over a time step Δt, the orientation update is given by:

$$q_{n+1} = q_n \cdot \exp\left(\frac{1}{2}\Delta t\,(0, \boldsymbol{\omega}_{n+1})\right).$$

The quaternion exponential is defined as:

$$\exp\left(\frac{1}{2}(0, \boldsymbol{\omega})\Delta t\right) = \left(\cos\left(\frac{\theta}{2}\right),\ \sin\left(\frac{\theta}{2}\right)\mathbf{u}\right),$$

with $\theta = \|\boldsymbol{\omega}\|\Delta t$, and $\mathbf{u} = \boldsymbol{\omega}/\|\boldsymbol{\omega}\|$ when $\boldsymbol{\omega} \neq 0$.

This integration rule preserves the unit norm of the quaternion and guarantees a valid rotation.

Remark 10.1. *The exponential of a general quaternion $q = (q_0, \mathbf{q}) \in \mathbb{H}$ is given by:*

$$\exp(q) = e^{q_0}\left(\cos\|\mathbf{q}\|,\ \sin\|\mathbf{q}\| \cdot \frac{\mathbf{q}}{\|\mathbf{q}\|}\right), \quad \textit{for } \mathbf{q} \neq \mathbf{0}.$$

In the case of a pure quaternion $(0, \boldsymbol{\omega})$*, this simplifies to:*

$$\exp\left((0, \boldsymbol{\omega})\right) = \left(\cos\|\boldsymbol{\omega}\|,\ \sin\|\boldsymbol{\omega}\| \cdot \frac{\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|}\right).$$

This is the form used in the quaternion time integration update, where the exponential encodes a rotation by angle $\theta = \|\boldsymbol{\omega}\|\Delta t$ about the axis $\boldsymbol{\omega}/\|\boldsymbol{\omega}\|$.

The angular velocity may also be represented in the body (material) frame as $\boldsymbol{\Omega}(t) \in \mathbb{R}^3$, and the two are related via the rotation matrix $Q(t) \in \mathrm{SO}(3)$ associated with $q(t)$:

$$\boldsymbol{\omega}(t) = Q(t)\boldsymbol{\Omega}(t), \quad \text{or equivalently,} \quad \boldsymbol{\Omega}(t) = Q(t)^\top \boldsymbol{\omega}(t).$$

This relationship is essential when converting between world-frame simulation results and body-frame expressions (e.g., for applying torque or computing inertial quantities).

Quaternion Rotation Representation A rotation by angle θ about a unit axis $\mathbf{u} \in \mathbb{R}^3$ is represented by the unit quaternion:

$$q = \left(\cos\left(\frac{\theta}{2}\right),\ \sin\left(\frac{\theta}{2}\right)\mathbf{u}\right).$$

Its conjugate (inverse rotation) is:

$$q^* = \left(\cos\left(\frac{\theta}{2}\right),\ -\sin\left(\frac{\theta}{2}\right)\mathbf{u}\right).$$

Pure Quaternion Identity For any $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, the product of two pure quaternions is:

$$(0, \mathbf{a}) \cdot (0, \mathbf{b}) = (-\mathbf{a} \cdot \mathbf{b},\ \mathbf{a} \times \mathbf{b}).$$

11 Discrete Variational Integrator for Rigid Body Motion

We now consider the discrete version of Hamilton's principle for rigid body motion, which includes both translation and rotation. The configuration of the system at time step n is represented by the center of mass position $c^n \in \mathbb{R}^3$ and the unit quaternion $q^n \in \mathbb{S}^3$ representing orientation.

We define the discrete action functional as a sum over a discrete Lagrangian:

$$S[\{c^n, q^n\}] = \sum_{n=1}^{N-1} L_d(c^{n-1}, q^{n-1}, c^n, q^n)\, \Delta t,$$

where Δt is the time step and L_d is the discrete Lagrangian approximating the action over one interval.

12 Forward Euler Method

The Forward Euler method is an explicit time integration scheme that updates quantities based on their current values. It is simple to implement and computationally inexpensive, though it may exhibit stability issues in stiff systems.

12.1 Rotation Update

Given the angular velocity vector $\omega(t)$, the rotation matrix $Q(t) \in \mathrm{SO}(3)$ evolves according to:

$$\dot{Q}(t) = \omega^{\times}(t) Q(t),$$

where $\omega^{\times}$ is the skew-symmetric matrix corresponding to ω. Using the Forward Euler method, the rotation matrix at the next time step is approximated by:

$$Q_{n+1} = Q_n + \Delta t \cdot \omega_n^{\times} Q_n.$$

Since this update does not preserve the orthonormality of Q, we re-orthonormalize Q_{n+1} after the update. A common approach is to use the QR decomposition:

$$Q_{n+1} \leftarrow \mathrm{orthonormalize}(Q_{n+1}).$$

This ensures that $Q_{n+1} \in \mathrm{SO}(3)$, maintaining the physical validity of the rotation.

12.2 Angular Velocity Update

The angular acceleration $\dot{\omega}$ is given by:

$$\dot{\omega} = I^{-1} \left(\tau - \omega \times (I\omega) \right),$$

where I is the inertia tensor and τ is the external torque. Using Forward Euler, the angular velocity update becomes:

$$\omega_{n+1} = \omega_n + \Delta t \cdot \dot{\omega}_n.$$

Substituting the expression for $\dot{\omega}_n$, we obtain:

$$\omega_{n+1} = \omega_n + \Delta t \cdot I^{-1} \left(\tau_n - \omega_n \times (I\omega_n) \right).$$

Remark 12.1. *The Forward Euler method is straightforward and efficient, making it suitable for initial exploration, visualization, or loosely coupled simulations. However, due to its explicit nature, it requires small time steps to ensure stability and accuracy, especially when modeling rapid rotational dynamics or systems with strong torques.*

13 Backward Euler

13.1 Governing Equation

$$\boldsymbol{\omega}\times = \dot{\mathbf{Q}}\mathbf{Q}^T \Longleftrightarrow \dot{\mathbf{Q}} = \boldsymbol{\omega} \times \mathbf{Q}$$
$$\mathbf{I}[\Omega_t] = \mathbf{Q}_n \mathbf{I}[\Omega_0] \mathbf{Q}_n^T$$
$$\boldsymbol{\tau} = \boldsymbol{\omega} \times \mathbf{I}[\Omega_t]\boldsymbol{\omega} + \mathbf{I}[\Omega_t]\dot{\boldsymbol{\omega}} \Longleftrightarrow \dot{\boldsymbol{\omega}} = \mathbf{I}[\boldsymbol{\Omega}_\mathbf{t}]^{-1} \left(\boldsymbol{\tau} - \boldsymbol{\omega} \times \mathbf{I}[\Omega_t]\boldsymbol{\omega}\right)$$

13.2 Rotation Update

The orientation update is given by:

$$\mathbf{Q}_{n+1} = \mathbf{Q}_n + \Delta t \cdot \dot{\mathbf{Q}}_{n+1}.$$

Substituting $\dot{\mathbf{Q}}_{n+1} = \boldsymbol{\omega}_{n+1} \times \mathbf{Q}_{n+1}$, we obtain:

$$\mathbf{Q}_{n+1} = \mathbf{Q}_n + \Delta t \left(\boldsymbol{\omega}_{n+1}^{\times} \mathbf{Q}_{n+1}\right).$$

Rearranging:

$$(\mathbf{I} - \Delta t \cdot \boldsymbol{\omega}_{n+1}^{\times})\mathbf{Q}_{n+1} = \mathbf{Q}_n.$$

Solving for $\mathbf{Q}_{n+1}$:

$$\mathbf{Q}_{n+1} = (\mathbf{I} - \Delta t \cdot \boldsymbol{\omega}_{n+1}^{\times})^{-1}\mathbf{Q}_n.$$

13.3 Inertia Update

The inertia tensor transforms as:

$$\mathbf{I}[\Omega_{t+1}] = \mathbf{Q}_{n+1}\mathbf{I}[\Omega_0]\mathbf{Q}_{n+1}^T.$$

Substituting the expression for $\mathbf{Q}_{n+1}$:

$$\mathbf{I}[\Omega_{t+1}] = \left((\mathbf{I} - \Delta t\boldsymbol{\omega}_{n+1}^{\times})^{-1}\mathbf{Q}_n\right)\mathbf{I}[\Omega_0]\left((\mathbf{I} - \Delta t\boldsymbol{\omega}_{n+1}^{\times})^{-1}\mathbf{Q}_n\right)^T.$$

Expanding:

$$\mathbf{I}[\Omega_{t+1}] = (\mathbf{I} - \Delta t\boldsymbol{\omega}_{n+1}^{\times})^{-1}\mathbf{Q}_n\mathbf{I}[\Omega_0]\mathbf{Q}_n^T(\mathbf{I} - \Delta t\boldsymbol{\omega}_{n+1}^{\times})^{-T}.$$

13.4 Torque and Angular Velocity Update

The angular velocity update follows:

$$\boldsymbol{\omega}_{n+1} = \boldsymbol{\omega}_n + \Delta t\dot{\boldsymbol{\omega}}_{n+1}.$$

Substituting the expression for angular acceleration:

$$\dot{\boldsymbol{\omega}}_{n+1} = \mathbf{I}[\boldsymbol{\Omega}_\mathbf{t}]^{-1}\left(\boldsymbol{\tau} - \boldsymbol{\omega}_{n+1} \times \mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_{n+1}\right),$$

we obtain:

$$\mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_{n+1} = \mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_n + \Delta t\left(\boldsymbol{\tau} - \boldsymbol{\omega}_{n+1} \times \mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_{n+1}\right).$$

Rearranging:

$$\mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_{n+1} + \Delta t\boldsymbol{\omega}_{n+1} \times \mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_{n+1} = \mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_n + \Delta t\boldsymbol{\tau}.$$

13.5 Backward Euler for Angular Velocity

To apply the Backward Euler method to solve for the angular velocity $\boldsymbol{\omega}_{n+1}$, we define the function:

$$F(\boldsymbol{\omega}_{n+1}) = \mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_{n+1} + \Delta t \boldsymbol{\omega}_{n+1} \times \mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_{n+1} - \mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_n - \Delta t \boldsymbol{\tau}.$$

Our goal is to solve for $\boldsymbol{\omega}_{n+1}$ by finding the root of $F(\boldsymbol{\omega}_{n+1}) = 0$.

13.6 Computing the Jacobian Matrix

To solve the nonlinear equation for angular velocity, we compute the Jacobian matrix:

$$\mathbf{J} = \frac{\partial F(\boldsymbol{\omega}_{n+1})}{\partial \boldsymbol{\omega}_{n+1}}.$$

This requires differentiating each term in $F(\boldsymbol{\omega}_{n+1})$.

13.6.1 Step 1: Define the Correction Matrix

The rotation update from earlier was:

$$\mathbf{Q}_{n+1} = (\mathbf{I} - \Delta t \cdot \boldsymbol{\omega}_{n+1}^{\times})^{-1}\mathbf{Q}_n.$$

Define the correction matrix:

$$\mathbf{C}(\boldsymbol{\omega}_{n+1}) = \mathbf{I} - \Delta t \boldsymbol{\omega}_{n+1}^{\times}.$$

The partial derivative of its components is given by:

$$\frac{\partial C_{ij}}{\partial \omega_{n+1,k}} = -\Delta t \epsilon_{ijk},$$

where ϵ_{ijk} is the Levi-Civita symbol:

$$\epsilon_{ijk} = \begin{cases} +1, & \text{if } (i,j,k) \text{ is an even permutation of } (1,2,3), \\ -1, & \text{if } (i,j,k) \text{ is an odd permutation of } (1,2,3), \\ 0, & \text{if any two indices are equal.} \end{cases}$$

13.6.2 Step 2: Compute the Derivative of $\mathbf{B}(\boldsymbol{\omega}_{n+1})$

Define the inverse of $\mathbf{C}(\boldsymbol{\omega}_{n+1})$ as:

$$\mathbf{B}(\boldsymbol{\omega}_{n+1}) = \mathbf{C}^{-1}(\boldsymbol{\omega}_{n+1}).$$

Applying the chain rule:

$$\frac{\partial B_{ij}}{\partial \omega_{n+1,k}} = \frac{\partial B_{ij}}{\partial C_{mn}} \frac{\partial C_{mn}}{\partial \omega_{n+1,k}}.$$

From the inverse relation:

$$C_{im} B_{mj} = \delta_{ij},$$

where the Kronecker delta is:

$$\delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

Differentiating both sides with respect to C_{pq},

$$\delta_{ip}\delta_{mq}B_{mj} + C_{im}\frac{\partial B_{mj}}{\partial C_{pq}} = 0.$$

Multiplying both sides by B_{ni}, we obtain:

$$B_{np}B_{qj} + \delta_{nm}\frac{\partial B_{mj}}{\partial C_{pq}} = 0.$$

Solving for the derivative:

$$\frac{\partial B_{mj}}{\partial C_{pq}} = -B_{mp}B_{qj}.$$

Thus:

$$\frac{\partial B_{ij}}{\partial C_{mn}} = -B_{im}B_{nj}.$$

13.6.3 Step 3: Compute $\frac{\partial B_{ij}}{\partial \omega_{n+1,k}}$

Using the previously computed derivatives:

$$\frac{\partial C_{mn}}{\partial \omega_{n+1,k}} = -\Delta t \epsilon_{mnk}.$$

Substituting into the chain rule:

$$\frac{\partial B_{ij}}{\partial \omega_{n+1,k}} = (-B_{im}B_{nj})(-\Delta t \epsilon_{mnk}) = \Delta t B_{im}B_{nj}\epsilon_{mnk}.$$

13.6.4 Computing the Derivative of $I[\Omega_{t+1}]$

The transformed inertia tensor at time $t+1$ is:

$$I[\Omega_{t+1}] \equiv A(\boldsymbol{\omega}_{n+1}) = B(\boldsymbol{\omega}_{n+1})Q_n I[\Omega_0]Q_n^T B^T(\boldsymbol{\omega}_{n+1}).$$

Differentiating:

$$\frac{\partial A_{ij}}{\partial \omega_{n+1,k}} = \Delta t B_{ia}B_{mb}\epsilon_{abk}Q_{n,mp}I[\Omega_0]_{pq}Q_{n,nq}B_{jn} + \Delta t B_{im}Q_{n,mp}I[\Omega_0]_{pq}Q_{n,nq}B_{ja}B_{nb}\epsilon_{abk}.$$

13.6.5 Computing the Jacobian J_{ik}

Recall that nonlinear function for the angular velocity update is:

$$F(\boldsymbol{\omega}_{n+1}) = A\boldsymbol{\omega}_{n+1} + \Delta t\, \boldsymbol{\omega}_{n+1} \times I[\Omega_{t+1}]\boldsymbol{\omega}_{n+1} - A\boldsymbol{\omega}_n - \Delta t \boldsymbol{\tau}.$$

Component-wise:

$$F_i(\boldsymbol{\omega}_{n+1}) = A_{ij}\omega_{n+1,j} + \Delta t \epsilon_{ilm}\omega_{n+1,l}A_{mj}\omega_{n+1,j} - A_{ij}\omega_{n,j} - \Delta t \tau_i.$$

The Jacobian is:

$$J_{ik} = \frac{\partial F_i}{\partial \omega_{n+1,k}}.$$

Differentiating each term:

1. First term:
$$\frac{\partial}{\partial \omega_{n+1,k}} \left(A_{ij}\omega_{n+1,j}\right) = A_{ij}\delta_{jk} + \frac{\partial A_{ij}}{\partial \omega_{n+1,k}}\omega_{n+1,j}.$$

2. Second term:
$$\Delta t \epsilon_{ilm} \left(\delta_{lk} A_{mj}\omega_{n+1,j} + \omega_{n+1,l}\frac{\partial A_{mj}}{\partial \omega_{n+1,k}}\omega_{n+1,j} + \omega_{n+1,l}A_{mj}\delta_{jk}\right).$$

The Jacobian matrix is given by:

$$J_{ik} = \frac{\partial A_{ij}}{\partial \omega_{n+1,k}}\omega_{n+1,j} + A_{ij}\delta_{jk} + \Delta t \epsilon_{ilm} \left(\delta_{lk} A_{mj}\omega_{n+1,j} + \omega_{n+1,l}\frac{\partial A_{mj}}{\partial \omega_{n+1,k}}\omega_{n+1,j} + \omega_{n+1,l}A_{mj}\delta_{jk}\right).$$

where the derivative of A_{ij} with respect to $\omega_{n+1,k}$ is given by:

$$\frac{\partial A_{ij}}{\partial \omega_{n+1,k}} = \Delta t B_{ia} B_{mb} \epsilon_{abk} Q_{n,mp} I[\Omega_0]_{pq} Q_{n,nq} B_{jn} + \Delta t B_{im} Q_{n,mp} I[\Omega_0]_{pq} Q_{n,nq} B_{ja} B_{nb} \epsilon_{abk}.$$

13.7 Newton's Iteration Steps

The Backward Euler update for angular velocity $\boldsymbol{\omega}_{n+1}$ is solved using Newton's method.

- **Step 1: Initialization** Set the initial guess:
$$\boldsymbol{\omega}_{n+1}^{(0)} = \boldsymbol{\omega}_n.$$

- **Step 2: Iterative Newton Updates** Repeat until convergence:

  1. Compute the function:
  $$F(\boldsymbol{\omega}_{n+1}^{(k)}) = \mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_{n+1}^{(k)} + \Delta t \boldsymbol{\omega}_{n+1}^{(k)} \times \mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_{n+1}^{(k)} - \mathbf{I}[\Omega_{t+1}]\boldsymbol{\omega}_n - \Delta t \boldsymbol{\tau}.$$

  2. Compute the Jacobian matrix:
  $$J = \frac{\partial F}{\partial \boldsymbol{\omega}_{n+1}^{(k)}}.$$

  3. Solve for the Newton step $\Delta\boldsymbol{\omega}$:
  $$J\Delta\boldsymbol{\omega} = -F(\boldsymbol{\omega}_{n+1}^{(k)}).$$

  4. Update the solution:
  $$\boldsymbol{\omega}_{n+1}^{(k+1)} = \boldsymbol{\omega}_{n+1}^{(k)} + \Delta\boldsymbol{\omega}.$$

- **Step 3: Convergence Check** The iteration stops when the Newton step satisfies:
$$\|F\| < \text{tol}.$$

14 Backward Euler with Newton's Second Law (Rigid Body Reduction)

14.1 Backward Euler Time Integration

Given position $\mathbf{x}^n$, velocity $\mathbf{v}^n$, and time step Δt, the implicit Backward Euler update is:

$$\frac{\mathbf{x}^{n+1} - \mathbf{x}^n}{\Delta t} = \mathbf{v}^{n+1}, \quad \mathbf{M}\frac{\mathbf{v}^{n+1} - \mathbf{v}^n}{\Delta t} = -\frac{\partial PE}{\partial \mathbf{x}}(\mathbf{x}^{n+1})$$

Combining the equations:

$$\mathbf{M}\left(\frac{\left(\frac{\mathbf{x}^{n+1}-\mathbf{x}^n}{\Delta t}\right) - \mathbf{v}^n}{\Delta t}\right) = -\frac{\partial PE}{\partial \mathbf{x}}(\mathbf{x}^{n+1})$$

Multiply both sides by Δt^2 and rearrange:

$$\mathbf{M}(\mathbf{x}^{n+1} - \hat{\mathbf{x}}^n) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{x}}(\mathbf{x}^{n+1}) = 0 \quad \Longleftrightarrow \quad \frac{\partial}{\partial \mathbf{x}}\Psi(\mathbf{x}) = 0$$

where

$$\hat{\mathbf{x}}^n := \mathbf{x}^n + \Delta t \mathbf{v}^n$$

and the energy-like functional is defined as:

$$\Psi(\mathbf{x}) := \frac{1}{2}(\mathbf{x} - \hat{\mathbf{x}}^n)^T \mathbf{M}(\mathbf{x} - \hat{\mathbf{x}}^n) + \Delta t^2 \cdot PE(\mathbf{x})$$

Thus, the update step is equivalent to solving the optimization problem:

$$\mathbf{x}^{n+1} = \arg\min_{\mathbf{x}} \Psi(\mathbf{x})$$

Note that in index notation:

$$\Psi(\mathbf{x}) = \frac{1}{2}(\mathbf{x}_{p\gamma} - \hat{\mathbf{x}}_{p\gamma})\mathbf{M}_{pq\gamma\epsilon}(\mathbf{x}_{q\epsilon} - \hat{\mathbf{x}}_{q\epsilon}) + \Delta t^2 \cdot PE(\mathbf{x})$$

where $\mathbf{M}$ is block diagonal, so that:

$$\mathbf{M}_{pq\gamma\epsilon} = m_p \, \delta_{pq} \, \delta_{\gamma\epsilon}$$

14.2 Gradient of the Energy Functional

Let

$$\mathbf{y}_{q\alpha}(\theta, \mathbf{c}) = \mathbf{R}_{\alpha\sigma}(\theta)\,(\mathbf{x}_{q\sigma} - \hat{\mathbf{x}}_\sigma) + \mathbf{c}_\alpha$$

where $\mathbf{c} \in \mathbb{R}^2$, $\theta \in \mathbb{R}$, and $\alpha = 1, 2$.

We now compute the gradient of the energy functional $\Psi(\mathbf{y})$ with respect to $\mathbf{y}_{p\delta}$, where:

$$\Psi(\mathbf{y}) = \frac{1}{2}(\mathbf{y}_{p\gamma} - \hat{\mathbf{x}}_{p\gamma})\mathbf{M}_{pq\gamma\epsilon}(\mathbf{y}_{q\epsilon} - \hat{\mathbf{x}}_{q\epsilon}) + \Delta t^2 \cdot PE(\mathbf{y})$$

$$\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}) = \frac{\partial}{\partial \mathbf{y}_{p\delta}} \left[\frac{1}{2}(\mathbf{y}_{p\gamma} - \hat{\mathbf{x}}_{p\gamma})\mathbf{M}_{pq\gamma\epsilon}(\mathbf{y}_{q\epsilon} - \hat{\mathbf{x}}_{q\epsilon}) + \Delta t^2 \cdot PE(\mathbf{y}) \right]$$
$$= \mathbf{M}_{pq\delta\epsilon}(\mathbf{y}_{q\epsilon} - \hat{\mathbf{x}}_{q\epsilon}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\delta}}(\mathbf{y})$$

Using the fact that $\mathbf{M}$ is block diagonal, i.e.,

$$\mathbf{M}_{pq\delta\epsilon} = M_{pq}\, \delta_{\delta\epsilon},$$

we get:

$$\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}) = M_{pq}(\mathbf{y}_{q\delta} - \hat{\mathbf{x}}_{q\delta}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}).$$

14.3 Rigid Body Reduction and Derivatives

We define the reduced energy functional as:

$$\hat{\Phi}(\theta, \mathbf{c}) := \Psi(\mathbf{y}(\theta, \mathbf{c}))$$

Our goal is to compute gradients of $\hat{\Phi}$ with respect to the rigid-body parameters $(\theta, \mathbf{c})$.

Translational Derivative By the chain rule:

$$\frac{\partial \hat{\Phi}}{\partial \mathbf{c}_\alpha} = \frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}(\theta, \mathbf{c})) \cdot \frac{\partial \mathbf{y}_{p\delta}}{\partial \mathbf{c}_\alpha}$$

Let $u_p = 1$, and assume the mass matrix is diagonal: $\mathbf{M}_{pq\delta\epsilon} = m_p \delta_{pq} \delta_{\delta\epsilon}$.
From earlier, we know the gradient of Ψ is:

$$\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} = m_p(\mathbf{y}_{p\delta} - \hat{\mathbf{x}}_{p\delta}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\delta}}$$

Next, recall the rigid-body motion:

$$\mathbf{y}_{p\delta} = \mathbf{R}_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) + \mathbf{c}_\delta \quad \Rightarrow \quad \frac{\partial \mathbf{y}_{p\delta}}{\partial \mathbf{c}_\alpha} = \delta_{\delta\alpha}$$

Substituting into the chain rule:

$$\frac{\partial \hat{\Phi}}{\partial \mathbf{c}_\alpha} = \sum_p \left[m_p(\mathbf{y}_{p\alpha} - \hat{\mathbf{x}}_{p\alpha}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\alpha}} \right] \cdot u_p$$

Since $u_p = 1$, we have:

$$\frac{\partial \hat{\Phi}}{\partial \mathbf{c}_\alpha} = \sum_p m_p(\underbrace{\mathbf{y}_{p\alpha}}_{\mathbf{R}_{\alpha\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) + \mathbf{c}_\alpha} - \hat{\mathbf{x}}_{p\alpha}) + \Delta t^2 \sum_p \frac{\partial PE}{\partial \mathbf{y}_{p\alpha}}$$

If $\hat{\mathbf{x}}_p = \hat{\mathbf{x}}$ for all p, and noting:

$$\sum_p m_p(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) = 0$$

then the mass-weighted rotation term vanishes:

$$\sum_p m_p \mathbf{R}_{\alpha\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) = \mathbf{R}_{\alpha\sigma}(\theta) \sum_p m_p(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) = 0$$

So the first term simplifies to:

$$\sum_p m_p(\mathbf{c}_\alpha - \hat{\mathbf{x}}_{p\alpha}) = \left(\sum_p m_p\right) \mathbf{c}_\alpha - \sum_p m_p \hat{\mathbf{x}}_{p\alpha}$$

Let $m := \sum_p m_p$. Then the translation gradient becomes:

$$\frac{\partial \hat{\Phi}}{\partial \mathbf{c}_\alpha} = m\mathbf{c}_\alpha - \sum_p m_p \hat{\mathbf{x}}_{p\alpha} + \Delta t^2 \sum_p \frac{\partial PE}{\partial \mathbf{y}_{p\alpha}}$$

Rotation Derivative Similarly, for the rotation parameter θ, we apply the chain rule:

$$\frac{\partial \hat{\Phi}}{\partial \theta} = \frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} \cdot \frac{\partial \mathbf{y}_{p\delta}}{\partial \theta}$$

To compute $\frac{\partial \mathbf{y}_{p\delta}}{\partial \theta}$, recall the rigid-body transformation:

$$\mathbf{y}_{p\delta} = \mathbf{R}_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) + \mathbf{c}_\delta$$

Therefore:

$$\frac{\partial \mathbf{y}_{p\delta}}{\partial \theta} = \frac{\partial \mathbf{R}_{\delta\sigma}}{\partial \theta}(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma)$$

Let $\mathbf{R}(\theta)$ be the standard 2D rotation matrix:

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad \Rightarrow \quad \mathbf{R}'(\theta) = \begin{bmatrix} -\sin\theta & -\cos\theta \\ \cos\theta & -\sin\theta \end{bmatrix}$$

Then the derivative becomes:

$$\frac{\partial \mathbf{y}_{p\delta}}{\partial \theta} = \mathbf{R}'_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma)$$

Substituting into the chain rule:

$$\frac{\partial \hat{\Phi}}{\partial \theta} = \sum_p \frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} \cdot \mathbf{R}'_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma)$$

Using the expression for $\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}}$ from before:

$$\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}) = M_{pq}(\mathbf{y}_{q\delta} - \hat{\mathbf{x}}_{q\delta}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}).$$

We arrive at:

$$\frac{\partial \hat{\Phi}}{\partial \theta} = \left[M_{pq}(\mathbf{y}_{p\delta} - \hat{\mathbf{x}}_{p\delta}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\delta}}\right] \cdot \mathbf{R}'_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma)$$